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Choosing where to sample next: adaptive designs for prevalence surveys

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Abstract. Surveys are expensive, and the value of a new survey location depends on what is already known. We propose an adaptive design that selects the next batch of locations to reduce the predictive variance where decisions are most uncertain. Using data from repeated surveys we compare the adaptive design with random and regular designs of the same size. The adaptive design reached the target precision with fewer locations in every scenario we considered, and the gains were largest when prevalence was close to the intervention threshold.

Keywords: adaptive sampling; geostatistical design; disease surveillance

1 Introduction

Disease control programmes plan their work district by district, but the data that inform those plans come from a limited number of survey locations. Between the locations we must predict, and every prediction carries uncertainty that should travel with it into the decision.

Geostatistical models treat the unobserved prevalence surface as a realisation of a stochastic process. The correlation between two locations decays with the distance between them at a rate set by the scale parameter of the correlation function. Locations close to a survey are predicted with more confidence than locations far from any survey, which is the behaviour a programme needs to see.

In this paper we describe the model, explain how it is fitted and checked, and show how its output can be summarised for decision makers. Section 2 introduces the data and Section 3 the model. Section 4 reports a simulation study, and Section 5 discusses limitations and extensions.

2 Data

The surveys recorded the number of people tested and the number who tested positive in each community, together with the geographical coordinates of the community and a set of environmental covariates derived from satellite images.

A mock first page made for the GPH slide template. The journal, authors and results are invented.